MODULE 1: CREATING AN ACTIVE LEARNING ENVIRONMENT

PDEV 6800
INTRODUCTION TO TEACHING AND LEARNING IN HIGHER EDUCATION

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TEACHING TEAM

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Responsible Sections : PDEV 6800X, 2025 Fall

Course Description

Time & Location

Mondays, 10:30 AM – 12:20 PM
W2-201

- The course aims to equip all full-time research postgraduate (RPg) students with **basic teaching skills before assuming teaching assistant duties** for the department.
- This course is **10-hour mandatory training course**. At the end of the course, GTAs should be able to (1) **facilitate teaching in tutorials and laboratory settings**; (2) **provide meaningful feedback** to their students; and (3) design an **active learning environment** to engage their students.

COURSE GRADE

- Upon completing the training, students will be assigned **instructional delivery in front of a group of students for a minimum of 30 minutes** by their respective College/Thrust.
- The 30-minute instructional delivery will be **graded based on faculty instructor's feedback, student evaluation,** and other criteria if deemed appropriate by the respective School/IPO.
- Performance in the 10-hour training and assessment in the required 30-minute session(s) will be **reflected in the transcript as an aggregated Pass/Fail grade.**

COURSE LEARNING OUTCOMES

On successful completion of the proposed course, students will be able to:	
CLO 1	Identify fundamental theories and good practices in teaching and learning.
CLO 2	Design appropriate active learning activities to engage students.
CLO 3	Apply constructive alignment in designing a learning sequence.
CLO 4	Demonstrate teaching and facilitation skills in different teaching settings.
CLO 5	Formulate constructive feedback to assist students as they progress in their learning.

COURSE SCHEDULE

Plan & Design	Grading & Feedback	Prepare & Deliver	Manage Canvas	Deliver Tutorials
Week 6	**Week 7**	**Week 8**	**Week 10**	**Week 12**
Creating an active learning environment	Effective ways to give meaningful feedback	Effective presentation and facilitation skills	Digital tools to facilitate teaching and learning	Micro-teaching to demonstrate an active learning strategy in tutorial

Note: Due to time constraints, the micro-teaching sessions will be held in Week 10 and Week 12. Specifically, **two groups will likely deliver their presentations in Week 10**, while the **remaining groups will conduct theirs in Week 12** (to be updated again if necessary).

Course Assessment

Assessment Methods	Course Learning Outcomes
Online (Canvas quizzes, interactive activities)	CLO-1, CLO-2, CLO-3, CLO-5
Face-to-face (Hands-on activities)	CLO-1, CLO-2, CLO-3, CLO-5
Group Written Task (Designing a lesson Plan)	CLO-1
Micro-teaching Demo	CLO-1, CLO-4

INTENDED LEARNING OUTCOMES FOR MODULE 1

- **ILO-1: Understand the theories of outcome-based teaching and learning**
- **ILO-2: Recognize the importance and necessity of creating an active learning environment**
- **ILO-3: Be familiar with common strategies and methods to implement active learning into teaching and learning**

ROADMAP FOR TODAY

The shift

From Teaching to Learning

Foundation

Outcome-Based Education (OBE)

The Core

What & Why of Active Learning?

The Toolkit

Principles & Best Practices

A workshop

Synthesis & QA

Before the roadmap, let's explore.....Why Active Learning?

1st CLASS ACTIVITY: THINK-PAIR-SHARE

What is your most memorable learning experience as a student?

THE TRADITIONAL LECTURE MODEL: A CRITIQUE

Information Transmission Model

Passive reception

Assumes a one-size-fits all pace

Limited feedback on student understanding

*The one who **does the work** does the **learning**.*

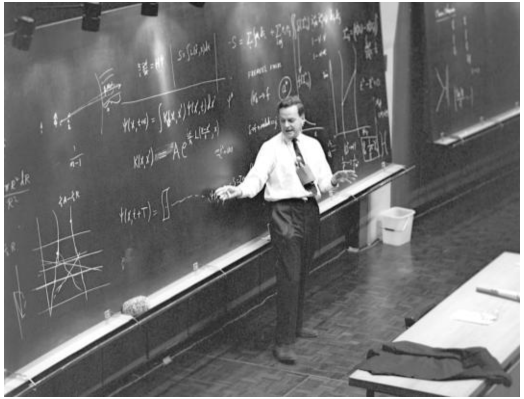
- Terry Doyle

The Paradigm Shift

01

ROADMAP ITEM 1: THE PARADIGM SHIFT

Teacher-centered



what the instructor teaches.

Student-centered

what the students can do.

Our role shifts from "**Sage on the Stage**" to "**Guide on the Side.**"

Outcome-Based Education (OBE)

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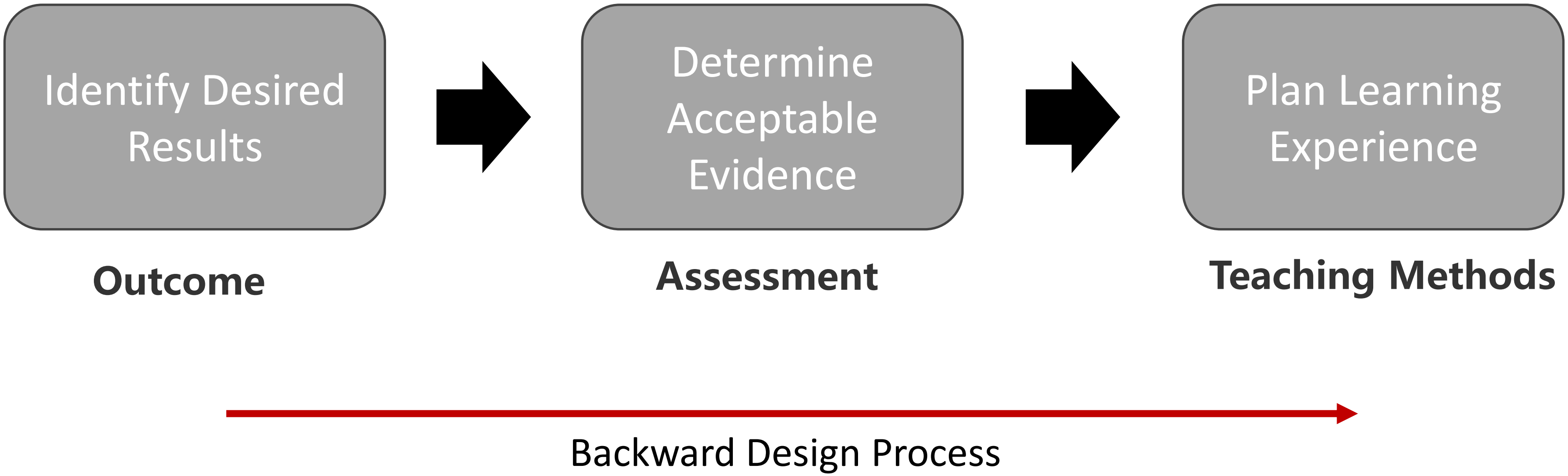
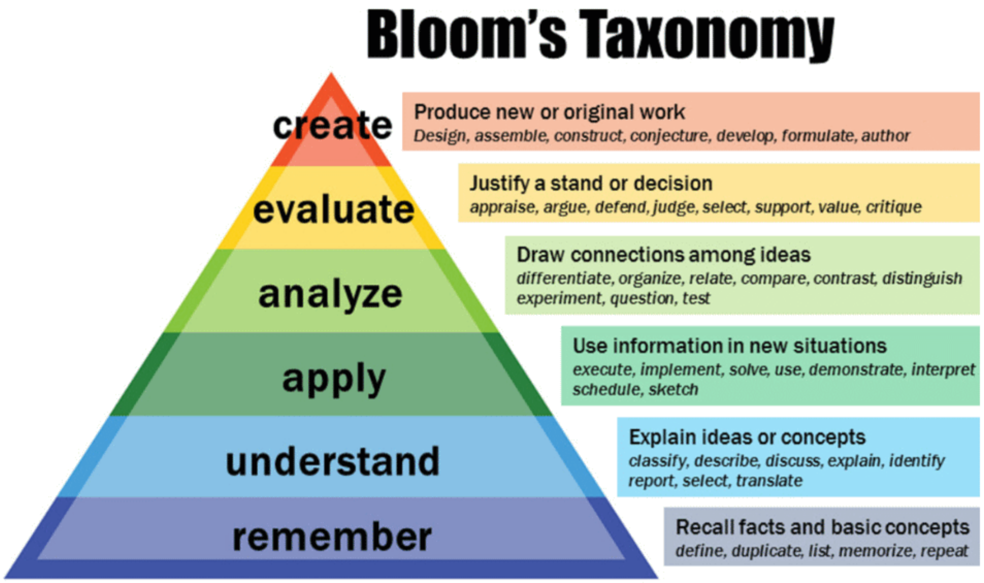
OUTCOME-BASED EDUCATION (OBE)

Definition: An educational theory that bases the design of all instruction and assessment around the clear, demonstrable **outcomes** students should be **able to perform upon completion**.



CRAFTING EFFECTIVE LEARNING OUTCOMES
(Identify Desirable Results)

Use **ACTION-ORIENTED VERBS** (Bloom's Taxonomy).



Formula: "By the end of this session/course, students will be able to [**Action Verb**] [**Specific Content/Skill**].“

Bad Example: *"Understand* **[Action verb]** *photosynthesis”* **[Content]**

Not specific enough.☹

Good Example: *" Explain* **[Action verb]** *the process of photosynthesis and diagram the light-dependent and light-independent reactions."***[Content]**

Content is clear and specific enough. ☺

DESIGN APPROPRIATE ASSESSMENTS
(Determining Acceptable Evidence)

Key Question: "How will I know if students have achieved the outcome?“

The assessment must measure the specific action verb in the outcome.

- Example Outcome: "Explain and diagram photosynthesis."
- ✓ Aligned Assessment: A short-answer question asking students to explain the process, and a task requiring them to diagram the reactions.
- ○ Misaligned Assessment: A multiple-choice question testing only the definition of "photosynthesis."

PLANNING LEARNING EXPERIENCES & INSTRUCTION (Teaching Methods)

Key Question: "What will students do to learn and practice?“

- This is where **Active Learning Strategies** are selected and implemented.
- The teaching method is chosen *because* it provides practice with the skills needed for the assessment and outcome.
- **Example:** To help students learn to *explain* and *diagram* photosynthesis, you could use a **Think-Pair-Share** to explain the steps, followed by a **collaborative whiteboard activity** to diagram the process.

The Core - Understanding Active Learning

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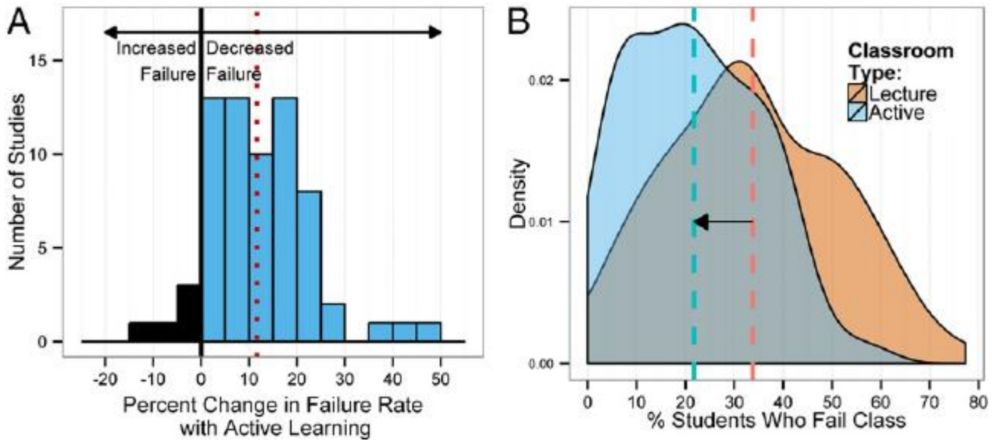
WHAT IS ACTIVE LEARNING?

Anything that involves students in **DOING** things and **THINKING** about the things they are doing. ***Bonwell & Eison***

Active Learning is when students are **MINDS-ON** and often **HANDS-ON** in meaningful learning activities.

Active Learning is **NOT** doing an activity for activity's sake; it's **purposeful engagement with the material**. – Must ensure student’s **ACHIEVING OUTCOMES**!

WHY ACTIVE LEARNING? THE EVIDENCE



- **Increases retention and understanding.**
- **Boosts student engagement and motivation.**
- **Develops critical thinking and problem-solving skills.**
- **Reduces achievement gaps for underrepresented students.**
- **Makes learning more visible to the instructor.**

Freeman, Scott, Sarah L. Eddy, Miles McDonough, Michelle K. Smith, Nnadozie Okoroafor, Hannah Jordt, and Mary Pat Wenderoth. "***Active learning increases student performance in science, engineering, and mathematics***." *Proceedings of the national academy of sciences* 111, no. 23 (2014): 8410-8415.

A LITTLE COGNITIVE SCIENCE BEHIND ACTIVE LEARNING

The Illusion of Competence

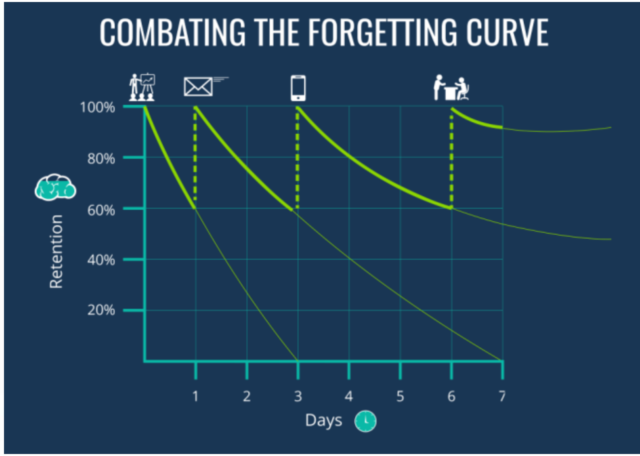
In a lecture, listening feels like learning, but it's often fleeting.

The Forgetting Curve

We forget rapidly without reinforcement.

Active Learning fights the curve

Retrieval Practice, Elaboration, Application.,



COMMON CONCERNS FROM TAs

"I have too much content to cover!"

- Active learning helps students learn core concepts better, making them more efficient. "Coverage" ≠ "Learning."

"Students will resist."

- Response: Explain the "Why," start small, and be consistent.

"It's chaotic and I'll lose control."

- Response: Clear instructions and structure are key. It's a structured engagement.

The Toolkit – Active Learning Strategies

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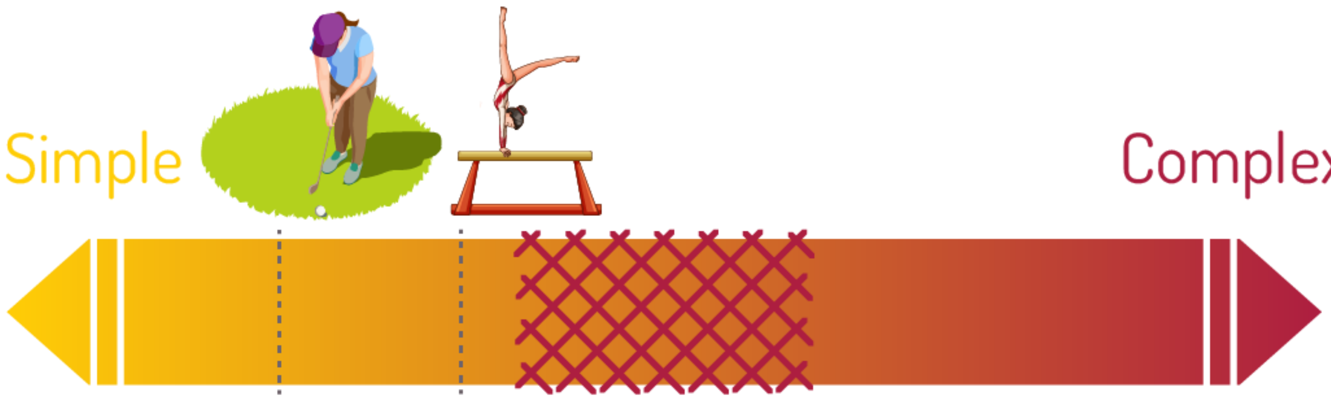
A SPECTRUM OF ACTIVE LEARNING STRATEGIES



LOW-STAKES / SIMPLE:

- Think-Pair-Share
- Polling (Mentimeter, Poll Everywhere)
- Minute Paper

COLLABORATIVE / MODERATE:

- Case Studies
- Jigsaw
- Structured Debates

HIGH-STAKES / COMPLEX:

- Problem-Based Learning
- Project-Based Learning
- Peer Instruction

THE ROLE OF THE TA IN AN ACTIVE CLASSROOM

You are a:

Facilitator, Guide, Coach, Resource.

Key Behaviors:

- Ask open-ended questions.
- Circulate and listen to group discussions.
- Provide timely and constructive feedback.
- Encourage a growth mindset.

The Jigsaw Workshop

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PREP FOR INTERACTIVE SESSION – JIGSAW ACTIVITY

We will now participate in a
Jigsaw Activity (moderate complexity active learning strategy)
to explore specific active learning techniques in detail.

- Step 1: Form Home Groups.
- Step 2: Expert Group Focus.
- Step 3: Become an Expert. (10 mins)
- Step 4: Teach Your Home Group. (10 mins)

我们现在将参与一个
拼图活动（中等复杂度的主动学习策略）
来详细探讨具体的主动学习技巧。

步骤 1： 组建原始小组。
步骤 2： 确定专家组重点。
步骤 3： 成为专家。 （10分钟）
步骤 4： 返回原始小组进行教学。（10分钟）

LET’S JIGSAW – STEP 1 & 2

Step 1: Form Home Groups. (8 mins)

- Form groups of 4 to 5 students with classmates near you. (3 mins)
- Assign a number from 1 to 5 to each student in the group. (3 mins)
- If your group has only 4 students, assign a number to be missing (1 to 2 minutes)

步骤 1：组建原始小组。（8分钟）

- 与附近的同学组成 4 到 5 人的小组。（3分钟）
- 给小组中的每位学生分配一个从 1 到 5 的编号。（3分钟）
- 如果你的小组只有 4 名学生，则缺少一个编号。（1-2分钟）

Step 2: Expert Group Focus. (2 mins)

- ➢ All #1s: Think-Pair-Share & Polling
- ➢ All #2s: Minute Paper & Muddiest Point
- ➢ All #3s: Case Studies
- ➢ All #4s: The Jigsaw Method
- ➢ All #5s: Peer Instruction

步骤 2：确定专家组重点。（2分钟）

- 所有 #1 号学生：思考-结对-分享 & 实时投票
- 所有 #2 号学生：一分钟报告 & 难点聚焦
- 所有 #3 号学生：案例研究
- 所有 #4 号学生：拼图教学法
- 所有 #5 号学生：同伴教学

LET’S JIGSAW – STEP 3

Step 3: Become an Expert. (10 mins)

- Move to your Expert Groups.
- Read the provided handout on your technique.
- Discuss: What is it? How does it work? What are its benefits? What are potential challenges and how to overcome them?
- Prepare a 2-minute summary for your Home Group.

步骤 3：成为专家。（10分钟）

- 移动到您所在的专家组。
- 阅读所提供的关于您分配到的教学技巧的资料。
- 讨论：它是什么？如何操作？有哪些好处？可能遇到哪些挑战以及如何克服？
- 为返回原始小组准备一个2分钟的总结。

LET’S JIGSAW – STEP 4

Step 4: Teach Your Home Group. (10 mins)

- Return to your Home Groups.
- Each "Expert" teaches their technique to the group (2 mins each).
- Group Task: As a Home Group, discuss and decide on one technique you would be most likely to try and why.

步骤 4：返回原始小组进行教学。（10分钟）

- 返回您所在的原始小组。
- 每位"专家"向小组成员讲解自己学习的教学技巧（每人2分钟）。
- 小组任务：作为原始小组，讨论并确定一个你们最可能尝试的技巧，并说明原因。

AFTER ACTIVITY DEBRIEF AND SHARING

- How did that feel as a learner?
- What was the cognitive demand compared to just listening?
- What logistical challenges did you notice?
- Let's hear from a few Home Groups: which technique might you prefer to try and why?

- 作为学习者，您感觉如何?
- 与单纯听讲相比，这种活动的认知需求有何不同?
- 您注意到了哪些流程上的挑战?
- 让我们听几个原始小组分享：你们可能更倾向于尝试哪种技巧，为什么?

Synthesis & Moving Forward

04

PRINCIPLES FOR CREATING AN ACTIVE LEARNING ENVIRONMENT

- **Start with Clear Outcomes:** What should students be able to *do*?
- **Explain the "Why":** Justify your methods to students.
- **Start Small:** Integrate one new technique at a time.
- **Plan and Structure Meticulously:** Clear instructions and timing are crucial.
- **Create a Supportive Climate:** It's okay to be wrong; we learn from mistakes.

- **从明确的学习成果开始**： 学生应该能够做什么？
- **解释"为什么"**： 向学生说明你采用这些方法的理由。
- **从小处着手**： 一次只尝试整合一种新技巧。
- **周密计划和构建**： 清晰的说明和时间安排至关重要。
- **创建支持性的氛围**： 犯错是可以的；我们能从错误中学习。

Leverage Technology for Active Learning!

利用技术促进主动学习！

ALIGNING OUTCOMES, ACTIVITY, AND ASSESSMENT

Outcome	Assessment	Teaching Methods
E.g.: “Students will be able to analyze a primary research article critically."	A short, guided analysis submitted at the end of class.	In-class, small groups analyze different sections of an article (Jigsaw method).

YOUR FIRST STEPS AS A TA

- **Observe:** How does your supervising professor create engagement?
- **Volunteer:** Propose to run a 15-minute active learning segment in a tutorial.
- **Reflect:** After a session, what worked? What could be improved?

Start with one technique from today.

观察： 您的指导教授是如何创造互动参与的？

主动提议： 建议在辅导课中负责一段15分钟的主动学习环节。

反思： 每次授课后，思考哪些做法有效？哪些可以改进？

从今天介绍的一种技巧开始尝试。

KEY TAKEAWAYS

- Your role is to **design** learning experiences, not just **deliver** content.
- Active learning is a powerful, evidence-based approach to **deepen understanding.**
- It is grounded in **clear learning outcomes**.
- **Start small**, be **purposeful**, and **reflect** on your practice.

- 您的角色是**设计**学习体验，而不仅仅是传授内容。
- 主动学习是一种强大的、基于证据的**深化理解**的方法。
- 它建立在**清晰的学习成果**基础之上。
- **从小**处着手，有**目的**地实践，并**反思您**的教学。

After-Class Assessment

Online Multiple-Choice Quiz & Sharing

Complete **the online quiz and sharing** to reinforce key concepts from today's session.

This assessment helps consolidate your understanding of outcome-based teaching and active learning principles.

THANK YOU! QUESTIONS?